Final Project: The Likelihood of Health Insurance Coverage based on Poverty Levels

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Abstract

The health insurance system in the United States has significant flaws, including high cost of coverage and difficulty of access, leaving millions uninsured or underinsured. This investigation is primarily focused on health insurance status, and more specifically, how an individual's poverty level is associated with the odds of having health insurance. Significant explanatory variables were selected using forward variable selection and the AIC, and the fitted logistic regression model with health coverage as a response was used to answer the research question. The analyses reveal that there is a statistically significant association between income level and the odds of being uninsured. Through this statistical analysis, we determine that the American insurance industry and health care system broadly has significant systemic issues that leave poor individuals in increasingly precarious positions.

Introduction

Access to healthcare in America is a contentious issue, with many Americans not receiving coverage for life-saving medical care. In a 2023 Census Bureau report, researchers determined that for all age groups uninsured rates are higher for those with incomes less than 100% of the poverty line (Keisler-Starkey and Bunch (2024)). HealthInsurance.org describes the federal poverty lines, stating that for the contiguous United States, the federal poverty line is approximately 31,000 for a family of four (Norris (n.d.)). Our report is interested in a more granular perspective on the impact of a person's poverty level on the odds that they are uninsured. We intend to characterize the significance of this pattern to showcase trends that may not be apparent in national data.

This report aims to understand how socioeconomic conditions, in terms of an individual's poverty level, are associated with the probability of having health insurance, before and after adjusting for other important predictors of an individual having health insurance coverage.

Data Description & Exploratory Data Analysis

The original dataset is from the Integrated Public Use Microdata Service, and it contains data collected by various national agencies such as the Census Bureau and the Bureau of Labor Statistics (Blewett et al. (2024)). This data comes from Americans across the contiguous United States, which represents the broader population that is our target of statistical inference. We are interested in the poverty level of an individual as an explanatory variable to the response variable, coverage status. The poverty level is a continuous numerical number and is defined as a percentage of the poverty line. Note that there are a small number of legitimate negative numbers as the poverty level that represent a net loss of income, for example through business losses. Coverage status is a binary variable, and indicates whether an individual has a form of health insurance or not.

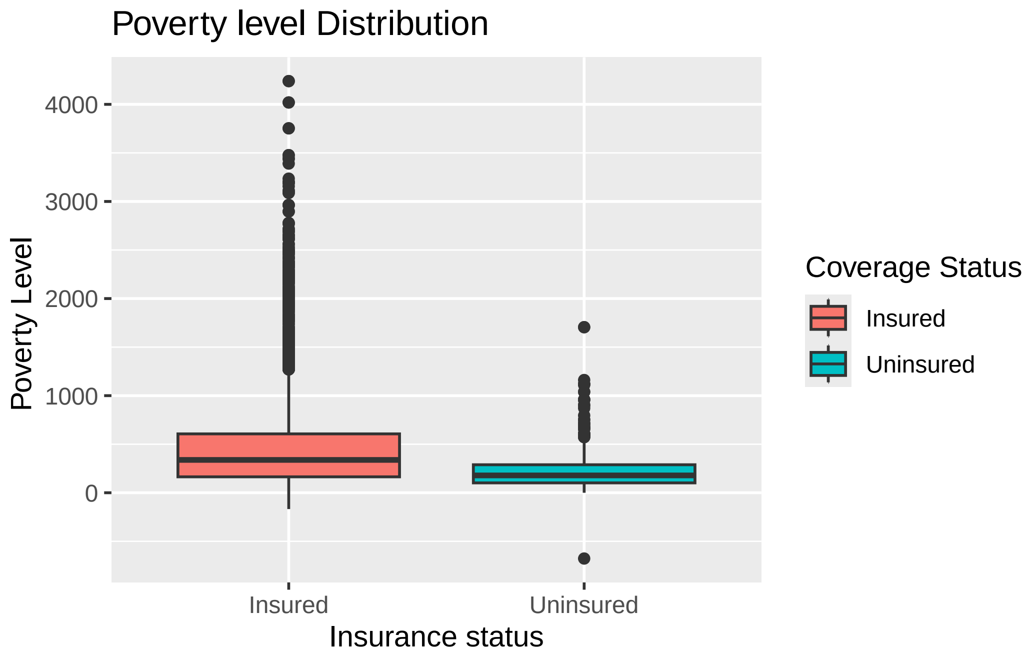
From the data that we extracted from Integrated Public Use Micro-data System (IPUMS) Health Surveys census data repository, we selected the variables HINOTCOV, POVLEV, AGE, SEX, MARSTAT, RACEA, YRSINUSG, USBORN, HEALTH, and POVLEV. These variables include information on whether the individual has health insurance or not (HINOTCOV), the individual's income as a percentage of the poverty line (POVLEV), the individual's age (AGE), the sex of the individual (SEX), the marriage status of the individual (MARSTAT), the whether the individual is white or non-white (RACEA), the number of years the individual has lived in the United States (YRSINUSG), whether the individual was born in the United States or not (USBORN), and the individual's health status (HEALTH).

The data was further cleaned by filtering for only those identifying as Male and Female in the SEX variable, filtering for where Health was reported (where the HEALTH variable did not equal 0, "Not in Universe"), where the number of years lived in the United States YRSINUSG was known, where whether the individual was born in the US or not was known, and organized the RACEA levels to simply be white and nonwhite, by consolidating all nonwhite observations into one level. We reorganized the RACEA variable levels to be white and nonwhite as the proportion of individuals in the data who were white completely outnumbered all instances of other races. We releveled the RACEA variable for simplicity and for more balanced factor levels for race.

The table below describes the distribution of poverty levels for those who are either insured or uninsured. There are 726 individuals who are uninsured and 10555 individuals who are insured in our dataset, and 11150 were removed from the dataset. For those who are uninsured, the minimum, median, maximum, and mean poverty level are less than those same summary statistics for those who have insurance coverage.

Status	n	Min_Poverty	Median_Poverty	Max_Poverty	Mean_Poverty	SD
Uninsured	726	-677.31	178.60	1705.04	226.9147	205.5436
Insured	10555	-167.96	338.31	4239.44	447.6481	418.2151

The figure above describes the same poverty level distribution for those who are insured and uninsured.



Statistical Methods

The modeling process for finding an appropriate combination of the relevant explanatory variables for our response variable of interest HINOTCOV was conducted using a forward selection variable selection technique, using AIC as a metric for comparison. The model that our analysis uses is such that the explanatory variable of interest, POVLEV, must have been included, so our model includes the variables for the individual's income as a percentage of the poverty line (POVLEV), the individual's age (AGE), whether the individual was born in the United States or not (USBORN), the sex of the individual (SEX), the number of years the individual has lived in the United States (YRSINUSG), the whether the individual is white or non-white (RACEA), the marriage status of the individual (MARSTAT), and whether the individual lives in the Midwest, West, Northeast, and South (REGIONMEPS). Below is the population model for our logistic regression.

$$logit(\pi(POVLEV_i, AGE_i, USBORN_i, SEX_i, YRSINUSG_i, RACEA_i, MARSTATWidowed_i,$$

$$MARSTATDivorced_i, MARSTATSeparated_i, MARSTATNeverMarried_i, REGIONMEPSMidwest_i, REGI$$

$$+\beta_{12}(REGIONMEPSSouth_i)+\beta_{13}(REGIONMEPSMWest_i)) = \beta_0+\beta_1(POVLEV_i)+\beta_2(AGE_i)+\beta_3(USBOR$$

$$+\beta_5(YRSINUSG_i) + \beta_6(RACEA_i) + \beta_7(MARSTATWidowed_i) + \beta_8(MARSTATDivorced_i)$$

$$+\beta_9(MARSTATSeparated_i)+\beta_{10}(MARSTATNeverMarried_i)+\beta_{11}(REGIONMEPSMidwest_i)$$

$$+\beta_{12}(REGIONMEPSSouth_i) + \beta_{13}(REGIONMEPSMWest_i)$$

Let $HINOTCOV_i$ denote whether the $i-th$ individual has health insurance or not, $POVLEV_i$ denote the individual's income as a percentage of the poverty line, AGE_i denote the individual's age, $USBORN_i$ denote whether the individual was born in the United States or not, SEX_i denote the individual's sex, $YRSINUSG_i$ denote the number of years the individual has lived in the United

States, $RACEA_i$ denote whether the individual is white or non-white, $MARSTATWidowed_i$ denote whether the individual is widowed or not, $MARSTATDivorced_i$ denote whether the individual is divorced or not, $MARSTATSeparated$ indicate whether the individual's marriage status is separated or not, $MARSTATNeverMarried$ indicate whether the individual had never been married or not, $REGIONMEPSMidwest_i$ indicate whether the individual lives in the Midwest or not, $REGIONMEPSSouth_i$ indicate whether the individual lives in the South or not, and$REGIONMEPSMWest_i$ indicate whether the individual lives in the West or not.

The model that considers only the individual's income as a percentage of the poverty line variable to estimate the health insurance status of an individual produces the following parameter estimate, seen in the table below. This indicates the exponentiating value for the odds of not having health insurance given that a study participant is at a particular income level group.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-1.804675	0.059565	-30.2976	0
POVLEV	-0.002808	0.000193	-14.5603	0

	2.5 %	97.5 %
(Intercept)	0.1462989	0.1847799
POVLEV	0.9968117	0.9975655

For the simple logistic regression model relating poverty level and health insurance status, a 1% percent increase in an individual's income as a percentage of the poverty line was associated with an estimated 0.9971959 multiplicative change in the odds of being uninsured.

We are 95% confident that a 1% percent increase in an individual's income as a percentage of the poverty line was associated with between a 0.9968117 and 0.9975655 times change in the odds of being uninsured.

The table below indicates the estimates of our logistic regression parameters, and their accompanying standard errors as well as their respective statistical significance as measured by the p-values.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	3.232222	0.669094	4.8307	0.0000
POVLEV	-0.002712	0.000212	-12.7724	0.0000
AGE	-0.041350	0.003086	-13.4009	0.0000
USBORN	-0.194107	0.032990	-5.8838	0.0000
SEX	-0.631171	0.085237	-7.4049	0.0000
YRSINUSG	-0.106604	0.072086	-1.4788	0.1392
RACEA	0.510082	0.099675	5.1175	0.0000
MARSTATWidowed	-0.516616	0.256015	-2.0179	0.0436
MARSTATDivorced	0.182057	0.136982	1.3291	0.1838
MARSTATSeparated	0.569366	0.209173	2.7220	0.0065
MARSTATNever Married	0.116849	0.108728	1.0747	0.2825
REGIONMEPSMidwest	0.549249	0.192505	2.8532	0.0043
REGIONMEPSSouth	1.418235	0.169109	8.3865	0.0000
REGIONMEPSWest	0.496000	0.186622	2.6578	0.0079

	2.5 %	97.5 %
(Intercept)	6.7374102	93.0541306
POVLEV	0.9968681	0.9976982
AGE	0.9536495	0.9652577
USBORN	0.7724069	0.8791635
SEX	0.4498514	0.6283852
YRSINUSG	0.7811067	1.0364363
RACEA	1.3730947	2.0299019
MARSTATWidowed	0.3506890	0.9617680
MARSTATDivorced	0.9136329	1.5639008
MARSTATSeparated	1.1586842	2.6355714
MARSTATNever Married	0.9080493	1.3907926
REGIONMEPSMidwest	1.1957524	2.5473331
REGIONMEPSSouth	2.9978758	5.8253768
REGIONMEPSWest	1.1479186	2.3895943

Holding all other important predictors of health insurance coverage status constant, a 1% percent increase in an individual's income as a percentage of the poverty line was associated with an estimated 0.9972917 times change in the odds of not having health insurance.

We are 95% confident that a 1% percent increase in an individual's income as a percentage of the poverty line was associated with between a 0.9968681 and 0.9976982 times change in the odds of being uninsured, given other important predictors of health insurance coverage status.

Both of the 95% confidence intervals associated with income as percentage of the poverty line (before and after adjusting for other important predictors of having health insurance) do not include an odds ratio of 1 within their ranges. An odds ratio of 1 means that the odds of being uninsured does not change as an individual's income increases. This indicates that there is statistically significant evidence for the association between an individual's poverty level and their odds of being uninsured. If the odds ratio of 1 is not contained within the 95% confidence interval, we can consider it an implausible value for the true odds ratio of uninsurance associated with income increase at the alpha level of 0.05.

Discussion

Through this report, we aimed to find how the poverty level of an individual is associated with the odds of having health insurance. We also were interested in which variables are significant for predicting the odds of having health insurance. After adjusting for these important variables, we aimed to understand if the poverty level was still significantly associated with the odds of having health insurance. First we tested a simple model with the only predictor being the poverty level of an individual, and found that this predictor was statistically significant. We are 95% confident that a 1% percent increase in an individual's income as a percentage of the poverty line was associated with between a 0.9968117 and 0.9975655 times change in the odds of being uninsured, and we are 95% confident that a 1% percent increase in an individual's income as a percentage of the poverty line was associated with between a 0.9968681 and 0.9976982 times change in the odds of being uninsured *given other important predictors of health insurance coverage status.*

Then we performed an analysis of possible variables and adjusted the simple model with the following factors: the individual's age, whether the individual was born in the US or not, the individual's sex, the number of years the individual has lived in the US, weather the individual is white or non-white, and the marriage status of the individual. After adjusting for these factors, we determined that poverty level remained a statistically significant predictor of whether an individual was uninsured or not. Ultimately we found that poverty level is an important determinant of whether someone has health insurance coverage or not, indicating that the American insurance industry and health care system broadly has significant systemic issues that leave poor individuals in increasingly precarious positions.

Much of our variable selection process could be expanded to include variables that may be important in a domain specific setting. In the IPUMS dataset, there were many variables describing individuals' lives, including socioeconomic factors, health factors, and more. We chose a subset of these factors to perform variable selection on, but there may have been other factors that were missed that could have contributed significantly to our model. Additionally, in our analysis we simplified the racial factors to simply white and non-white, but disaggregating this factor could provide more granular information about how variables associate with each other.

Blewett, Lynn A., Julia A. Rivera Drew, Daniel Backman, et al. 2024. "IPUMS Health Surveys: Medical Expenditure Panel Survey, Version 2.4 [Dataset]." *IPUMS*, ahead of print. https://doi.org/10.18128/D071.V2.4.

Keisler-Starkey, K., and L. Bunch. 2024. "Health Insurance Coverage in the United States: 2023." *United States Census Bureau*, September.

Norris, L. n.d. *Federal Poverty Level (FPL) / Federal Poverty Guidelines*. Https://www.healthinsurance.org/glossary/federal-poverty-level/.